Lesson 2.1: Principles of layout

What you learn

- **Visual hierarchy.** Size, weight, colour and position tell the eye where to look first.
- **Negative space.** Empty space gives focus and calm.
- **Colour theory.** Use a limited palette with one accent. RGB is for screens; CMYK is for print, so convert and proof before printing.
- **Contrast.** Text must be readable. The checker below measures the WCAG contrast ratio.

Tools and resources (official sites, not checked live)

- Adobe colour

ASSIGNMENT

Practice for lesson 2.1

Business problem: Check a palette.

Steps

1. Run the tested contrast checker.
2. Test your text and background.
3. Fix any failing pair.

Expected output: All text pairs at 4.5:1 or better.

SOLUTION (ANSWER KEY)

PYTHON

```
def _lin(c):
    c = c / 255
    return c / 12.92 if c <= 0.03928 else ((c + 0.055) / 1.055) ** 2.4

def luminance(hex_colour):
    h = hex_colour.lstrip("#")
    r, g, b = (int(h[i:i + 2], 16) for i in (0, 2, 4))
    return 0.2126 * _lin(r) + 0.7152 * _lin(g) + 0.0722 * _lin(b)

def contrast_ratio(fg, bg):
    l1, l2 = sorted([luminance(fg), luminance(bg)], reverse=True)
    return round((l1 + 0.05) / (l2 + 0.05), 2)

def passes_aa(fg, bg, large_text=False):
    return contrast_ratio(fg, bg) >= (3.0 if large_text else 4.5)
```

Dark ink on white scores 16.92. A light grey on white fails the 4.5 test.

Check yourself

1. RGB or CMYK for print?
   Answer: CMYK.

2. What ratio does normal text need?
   Answer: 4.5:1.

Lesson 2.1 of the **Design and Creative** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.